

PART II

SHORT-HAUL TRANSPORTATION: BASIC MODELS

7. The capacitated vehicle routing problem

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7.1 INTRODUCTION

The Capacitated Vehicle Routing Problem (CVRP) is the basic version of an extensively studied family of combinatorial optimization problems that model short-haul transportation applications, where the demand of the customers is typically (much) smaller than the vehicle capacity so that several customers may be served in a vehicle route. The CVRP extends the well-known Traveling Salesperson Problem (TSP), which looks for the shortest route for a single uncapacitated vehicle visiting a set of customers. Travel costs are given in the TSP and the CVRP; each customer must be visited once, and vehicles start and end at a depot. The aim in both problems is to minimize the sum of travel costs. The extension from TSP to CVRP also involves caring about customers' demands and vehicle capacities. Hence, the CVRP involves solving two combinatorial optimization problems. One problem is assigning customers to vehicles so that each vehicle can feasibly load all the demand required to serve all the customers assigned to it. The other problem is determining the shortest route for each vehicle visiting its assigned customers. The CVRP aims to solve both combinatorial problems simultaneously. It is worth noting that each problem individually is challenging (i.e., it is *NP-hard* in complexity theory terminology). The first problem is known as the Bin Packing Problem (BPP), and the second problem coincides with solving a TSP for each vehicle. Motivated by this decomposition and complexity, several heuristic approaches have been proposed in the literature for the CVRP. See, e.g., Laporte and Vidal (2014) for a comprehensive survey on heuristic algorithms for the CVRP. This chapter focuses on mathematical formulations that may be used to design exact methods for solving the CVRP.

7.2 PROBLEM DEFINITION AND NOTATION

In this section, we formally define the CVRP. Without loss of generality, we use the standard assumption that the customers correspond to deliveries, as stated originally in Dantzig and Ramser (1959). The demands are deterministic, known in advance, and assumed to be served in one visit (i.e., no split). The travel costs are also known in advance and assumed to be asymmetric (i.e., the cost of going between two points depends on which one is the origin and destination). Although several papers in the literature consider the case in which costs are symmetric, we consider asymmetric costs for clarity of exposition and notation, as well as for greater generality. The vehicles are identical and based at a single depot, and only capacity restrictions are imposed. The objective is to minimize the total travel cost to serve all the customers. Thus, in the basic definition, we neither minimize the number of vehicle routes in the solution nor limit the travel time of each vehicle route.

The CVRP may be described as the following problem in graph theory. Let $G = (V, A)$ be a complete and oriented graph where $V = \{0, 1, \dots, n\}$ is the vertex set and A is the arc set. Vertex 0 corresponds to the *depot* and the vertices in $N = \{1, \dots, n\}$ correspond to the *customers*. In the following, we generally identify an arc by a single index $a \in A$, and whenever required, we may use its endpoints vertices i and j explicitly to identify the arc $a = (i, j)$.

Let $c_a = c_{ij}$ be a non-negative *cost* associated with each arc $a = (i, j) \in A$, representing the *travel cost* to go from vertex i to vertex j . We exclude loop arcs (i, i) from A as they are not part of a CVRP solution. Given a vertex i , let $\delta^+(i)$ denote the so-called *forward star* of i , defined as the set of vertices j such that arc $(i, j) \in A$, i.e., the vertices that are directly reachable from i . Similarly, let $\delta^-(i)$ denote the *backward star* of vertex i , defined as the set of vertices j such that arc $(j, i) \in A$, i.e., the vertices from which i is directly reachable.

Extending this notation, given a subset $S \subset V$, we use $\delta^+(S)$ to represent all the arcs with origin in S and destination in $V \setminus S$. Similarly, $\delta^-(S)$ represents all the arcs with origin in $V \setminus S$ and destination in S , and we use $A(S)$ to represent all the arcs with both origin and destination in S .

While in general, an arc in A is represented by the two indices (i, j) determining the tail and the head of that arc, in this chapter, we represent an arc with one index a when possible, thus prioritizing notation simplicity in our exposition. In this line, an arc's cost may be written as c_{ij} or as c_a , depending on the context.

When the travel costs conform to a symmetric matrix, it may be more convenient to consider an undirected graph $G = (V, E)$ where each pair of arcs (i, j) and (j, i) is replaced by an edge $e = \{i, j\}$. Similarly to the notation introduced before, for a subset $S \subset V$, the edge subset $\delta(S)$ denotes all edges with one vertex in S and the other vertex in $V \setminus S$, and $E(S)$ denotes all the edges with both vertices in S . For simplicity, we write $\delta(i)$ instead of $\delta(\{i\})$ for each $i \in V$.

In several practical cases, the cost matrix satisfies the *triangle inequality*, $c_{ik} + c_{kj} \geq c_{ij}$ for all $i, j, k \in V$. In other words, any path between two vertices i and j , which also visits other vertices, always has a cost that is not smaller than the direct arc (i, j) . Among other circumstances, such a property is satisfied when the cost of each arc $(i, j) \in A$ is equal to the cost of the shortest path from i to j . Another example where the triangle inequality holds occurs when the cost matrix is computed using the Euclidean distances between Cartesian coordinates of the customer locations in a space. Observe that the frequently performed rounding to the nearest integer of the real-valued distance may cause a violation of the triangle inequality, whereas this does not happen if the real-valued numbers are rounded up. Some specialized algorithms can exploit the presence of the triangle inequality, and it can be artificially obtained by adding a suitably large positive quantity M to the cost of each arc. However, the drastic distortion of the metric induced by this operation may produce very poor lower and upper bounds with respect to those corresponding to the original costs. This chapter does not make any assumption about the travel cost matrix.

Each customer $i \in N$ is associated with a known positive *demand* q_i to be delivered. Since each customer must be visited once, we assume $q_i \leq Q$ for each $i \in N$. Also, when an arc (i, j) connects two customers i and j with $q_i + q_j > Q$, then this arc can be removed from the arc set A since it may never belong to a feasible solution. As a convention, one can assume that the depot has a fictitious demand $q_0 = 0$. Given a subset $S \subseteq N$, let $q(S) = \sum_{i \in S} q_i$ denote the total demand of the customers in S .

Let us consider a homogeneous fleet of identical vehicles initially available at the depot. Let m be the number of vehicles, $K = \{1, \dots, m\}$ be the vehicle set, and Q be the capacity of each vehicle. We assume that $q(N) \leq m \cdot Q$ increases the chance of obtaining a feasible solution. We assume that each vehicle performs exactly one route and that m is not smaller than the minimum number m^* of vehicles needed to serve the demand of all customers. The value m^* may be determined by solving the BPP associated with the CVRP, calling for the determination of the minimum number of bins, each with capacity Q , required to load all the n items, each with non-negative weight q_i for $i \in N$. Even if BPP is NP-hard in the strong sense, instances with hundreds of items can be solved optimally in a short computing time (see, e.g., Martello and Toth, 1990). Given a set $S \subseteq N$, we denote by $r(S)$ the minimum number of vehicles needed to serve the demand of all the customers in S , i.e., the optimal solution value of the BPP associated with item set S . Thus, $m^* = r(N)$. A trivial upper bound of $r(S)$ is the so-called continuous lower bound, given by $\lceil q(S)/Q \rceil$.

The CVRP consists of finding a collection of *circuits* (each corresponding to a vehicle route) with minimum cost (defined as the sum of the costs of the arcs belonging to the circuits) such that:

- each circuit visits the depot;
- each customer is visited by exactly one circuit;
- the sum of the demands of the customers visited by a circuit does not exceed the vehicle capacity Q ;
- there should be exactly m circuits in the collection.

Several variants of this basic definition of CVRP have been considered in the literature. First, when the number m of available vehicles is greater than m^* , it may be possible to relax (iv) and leave some vehicles unused, thus requiring determining *at most* m circuits. In this case, fixed costs are often associated with using the vehicles and added to the objective function in a weighted two-criteria objective function (see Chapter 8 for details). Another frequently considered variant arises when the fleet of vehicles is heterogeneous, i.e., each vehicle k has a capacity Q^k for $k \in K$. In some situations, routes containing only one customer may not be allowed.

In the next section, we discuss how some models for the basic CVRP can be adapted to consider these additional features. However, these extensions are also discussed in more detail in other chapters of this book (see, e.g., Chapters 6, 8, and 19).

A quite popular variant of CVRP is the so-called *Distance-Constrained VRP*, where for each route, the capacity constraint is replaced by a maximum length (or time) constraint. In particular, a non-negative *length* t_{ij} is associated with each arc $(i, j) \in A$, and the total length of the arcs of each route cannot exceed the given maximum route length T . If the vehicles differ, then the maximum route lengths are T_k for $k = 1, \dots, m$. Moreover, when arc lengths represent travel times, a *service time* s_i may be associated with each customer i , denoting the time period for which the vehicle must stop at its location. Alternatively, the service times can be added to the travel times of the arcs, i.e., by defining $t_{ij} = t'_{ij} + s_i / 2 + s_j / 2$, for each arc (i, j) , where t'_{ij} is the original travel time of arc (i, j) .

7.3 BASIC MODELS FOR THE VRP

In this section, we present the most popular mathematical programming formulations used to model the CVRP described in the previous section. We present the formulations grouped according to whether they are compact models (i.e., models with a number of variables and constraints polynomial in n and m). The compact models are *arc based*, meaning that binary variables decide whether an arc of the graph is part of a vehicle route in an optimal solution. These binary variables are also suitable for formulations with an exponential number of constraints managed with cutting-plane approaches. The models with an exponential number of variables are *route based*, meaning that binary variables decide whether a route is selected for a vehicle in an optimal solution. They are also called set-partitioning formulations and require appropriate column-generation techniques.

7.3.1 Compact Formulations

The compact models use an integer variable associated with each arc of the graph to count the number of times such an arc is traversed by a vehicle. They are particularly suited for cases in which the cost of the solution can be expressed as the sum of the costs associated with the arcs, and when the most relevant constraints concern the direct transition between the customers within the route, so they can be effectively modeled through an appropriate definition of the arc set and of the arc costs. On the other hand, these arc-based formulations cannot be used to handle many practical issues, e.g., when the cost of a solution depends on the overall vertex sequence.

The most intuitive approach for someone familiar with the TSP literature uses a binary variable x_a^k for each arc $a \in A$ and each vehicle $k \in K$ to determine whether a is traversed by k in an optimal solution. Although not necessary, it could simplify notation to have another set of binary variables y_i^k taking value 1 if customer i is served by vehicle k in an optimal solution and 0 otherwise. Then, a model for the CVRP based on these variables is as follows:

$$\min \sum_{a \in A} c_a \sum_{k \in K} x_a^k \quad (7.1)$$

subject to

$$\sum_{k \in K} y_i^k = 1 \quad \forall i \in N \quad (7.2)$$

$$\sum_{a \in \delta^+(i)} x_a^k = \sum_{a \in \delta^-(i)} x_a^k = y_i^k \quad \forall i \in N, k \in K \quad (7.3)$$

$$\sum_{a \in \delta^+(0)} x_a^k = \sum_{a \in \delta^-(0)} x_a^k = 1 \quad \forall i \in N, k \in K \quad (7.4)$$

$$\sum_{i \in N} q_i y_i^k \leq Q \quad \forall k \in K \quad (7.5)$$

$$[\text{no subtours in } N] \quad (7.6)$$

$$y_i^k \in \{0,1\} \quad \forall i \in N, k \in K \quad (7.7)$$

$$x_a^k \in \{0,1\} \quad \forall a \in A, k \in K. \quad (7.8)$$

The formulation was given by Golden et al. (1977) and is known in the literature as the three-index flow formulation for the three indices in the variables X_{ij}^k . Equations (7.2)–(7.4) impose that each customer is visited exactly once and each vehicle is used. Constraints (7.5) are the capacity restrictions for each vehicle k . Constraints (7.6) enforce that each customer is connected to the depot. The TSP literature shows different families of linear inequalities that can be used for this purpose, such as the so-called subtour elimination constraints described in Section 7.3.2.

Formulation (7.1)–(7.8) has the advantage that the vehicle capacity limitation is easily modeled through m inequalities in (7.5). These constraints even allow us to deal with heterogeneous fleets of vehicles by possibly defining a different capacity Q^k for each vehicle k . Also, the formulation allows a different cost c_a^k for traversing each arc a , depending on the vehicle. However, when all vehicles are identical (as assumed in this chapter), the formulation has the major disadvantage of representing a feasible collection of routes through many equivalent sets of values for the variables (one for each permutation of the indices in K). This high symmetry of the solution space creates many numerical issues for mixed integer linear solvers. Therefore, it is an undesired formulation also because we can easily eliminate it by using *one* variable for each arc instead of m variables.

Let us consider one variable to decide whether a will be routed by a vehicle, no matter which one, for each arc a of the graph. More precisely, let a variable x_a assume value 1 if arc $a \in A$ belongs to an optimal solution, and the value 0 otherwise. Then, a basic mathematical formulation could look like this:

$$\min \sum_{a \in A} c_a x_a \quad (7.9)$$

subject to

$$\sum_{a \in \delta^+(i)} x_a = \sum_{a \in \delta^-(i)} x_a = 1 \quad \forall i \in N \quad (7.10)$$

$$\sum_{a \in \delta^+(0)} x_a = \sum_{a \in \delta^-(0)} x_a = m \quad (7.11)$$

$$[\text{vehicle capacity limits}] \quad (7.12)$$

$$[\text{no subtours in } N] \quad (7.13)$$

$$x_a \in \{0,1\} \quad \forall a \in A. \quad (7.14)$$

The *in-degree* and *out-degree* constraints (7.10) impose that exactly one arc enters and leaves each vertex associated with a customer. Analogously, constraints (7.11) impose the degree requirements for the depot vertex. Note that one arbitrary equation among the $2|V|$ constraints (7.10) is actually implied by the remaining $2|V|-1$ ones; hence, it can be removed. The key issue is to model constraints (7.12) and (7.13) as a set of linear inequalities. Different procedures lead to different CVRP formulations, some described in this chapter. In contrast to the first compact formulation, each alternative formulation is known in the literature as a two-index flow formulation for the CVRP due to the two indices in the variables x_{ij} . In this section, we describe some models with a polynomial number of inequalities, leaving the models that include an exponential number of inequalities for the next section.

Miller–Tucker–Zemlin formulations: A first compact two-index model for the CVRP was given in Christofides et al. (1979), adapting the model proposed in Miller et al. (1960) for the TSP. It consists of adding a continuous variable u_i representing the amount of product delivered after visiting customer i , and then the model is the two-index formulation (7.9)–(7.14) with (7.12) and (7.13) replaced by:

$$u_j \geq u_i + q_j - Q(1 - x_{ij}) \quad \forall (i, j) \in A \mid i, j \in N \quad (7.15)$$

$$q_i \leq u_i \leq Q \quad \forall i \in N. \quad (7.16)$$

The formulation is referred to as MTZ1 in the following. It is easy to see that constraints (7.15) and (7.16) impose both the capacity and the connectivity requirements of CVRP. Indeed, when $x_{ij} = 0$ constraint (7.15) is not binding since $u_i \leq Q$ and $u_j \geq q_j$, whereas when $x_{ij} = 1$ they impose that $u_j \geq u_i + q_j$. It is worth observing that (7.15) does not guarantee $u_j = u_i + q_j$ when $x_{ij} = 1$, which could be forced by also including the inequalities $u_j \leq u_i + q_j + Q(1 - x_{ij})$ for all $(i, j) \in A$ with $i, j \in N$. Still, inequalities (7.15) and (7.16) are enough to certify that x defines a feasible CVRP solution.

Inequalities (7.15) and (7.16) also eliminate subtours disconnected from the depot because $q_j > 0$ for all $j \in N$. Desrochers and Laporte (1991) suggested the stronger inequalities:

$$u_j \geq u_i + q_j - Q(1 - x_{ij}) + (Q - q_i - q_j)x_{ji} \quad \forall (i, j) \in A \mid i, j \in N \quad (7.17)$$

$$q_i + \sum_{j \in N \setminus \{i\}} q_j x_{ji} \leq u_i \leq Q - \sum_{j \in N \setminus \{i\}} q_j x_{ij} \quad \forall i \in N. \quad (7.18)$$

Note that Q is used as a big coefficient in (7.15) to deactivate the inequality when $x_{ij} = 0$. The strength of (7.17) lies in reducing this coefficient and imposing $u_j \geq u_i - q_i$ instead of the

weaker inequality $u_j \geq u_i + q_j - Q$ when $x_{ji} = 1$, a particular case of $x_{ij} = 0$. The formulation (7.9)–(7.14) with (7.17)–(7.18) instead of (7.12)–(7.13) is referred to as MTZ2.

Formulations MTZ1 and MTZ2 can easily be modified to deal with a heterogeneous fleet of vehicles. Indeed, let z_i^k be a binary variable assuming value 1 when i is the last customer served by vehicle k . Then a formulation for the CVRP where vehicle k has capacity Q^k is (7.9)–(7.14) with (7.12) and (7.13) replaced by (7.15) and:

$$\begin{aligned} \sum_{i \in N} z_i^k &= 1 & \forall k \in K \\ \sum_{k \in K} z_i^k &= x_{i0} & \forall i \in N \\ z_i^k &\in \{0, 1\} & \forall i \in N, k \in K \\ q_i \leq u_i &\leq \sum_{k \in K} Q^k z_i^k + Q(1 - x_{i0}) & \forall i \in N, \end{aligned}$$

where $Q = \max\{Q^k \mid k \in K\}$. This system becomes inequalities (7.16) when all vehicles have identical capacity Q .

The last inequality can be slightly strengthened by introducing an additional set of binary 2-index variables s_i^k representing whether customer i is visited by vehicle k , but i is not the last customer visited by k . Find details in the compact formulation in [Section 8.4](#).

Single-commodity flow formulations: The previous model forces the transport and delivery of a single-commodity product from the depot to the customer by using one variable for each vertex. This can be seen as a flow and, therefore, can alternatively be represented by a variable for each arc, leading to the compact model presented in Gavish and Graves (1979). It uses a flow variable f_a for each arc a representing the vehicle load when traversing a . Then, the model is the two-index flow formulation (7.9)–(7.14) with (7.12) and (7.13) replaced by:

$$\sum_{a \in \delta^-(i)} f_a - \sum_{a \in \delta^+(i)} f_a = q_i \quad \forall i \in N \quad (7.19)$$

$$f_a \geq 0 \quad \forall a \in A \quad (7.20)$$

$$f_{0i} \leq Qx_{0i} \quad \forall i \in N. \quad (7.21)$$

Our assumption that a vehicle delivers when serving a customer makes inequalities (7.21) sufficient to guarantee the vehicle capacity limitation. We could either force that each vehicle leaves full from the depot (i.e., $f_{0i} = Qx_{0i}$) or arrives empty at the depot (i.e., $f_{0i} = 0$), but this does not affect the LP relaxation. A stronger formulation is obtained when (7.20)–(7.21) is replaced by the larger family of inequalities:

$$0 \leq f_a \leq Qx_a \quad \forall a \in A. \quad (7.22)$$

This strengthening leads to the so-called *single commodity flow formulation* for the CVRP, referred to here as SCF1a. Hoffman's Circulation Theorem (see Hoffman, 1960) proves that

the LP relaxation of the new formulation coincides with that of formulation (7.9)–(7.14) where (7.12) and (7.13) are replaced by the fractional capacity-cut inequalities (7.42) detailed in the next section.

One can improve the formulation SCF1a by observing that the load of the vehicle through an arc (i, j) is at least q_j if $j \in N$ and at most $Q - q_i$ if $i \in N$, whenever the vehicle traverses that arc. In a linear format, this means that the inequalities (7.20)–(7.21) can be replaced by the stronger inequalities:

$$q_j x_{ij} \leq f_{ij} \leq (Q - q_i) x_{ij} \quad \forall (i, j) \in A, \quad (7.23)$$

where $q_0 = 0$. We denote this new formulation as SCF1b.

Also, these models can easily be modified to deal with a heterogeneous fleet of vehicles. Indeed, let z_i^k be a binary variable assuming value 1 when i is the first customer served by vehicle k . Then a formulation for the CVRP where vehicle k has capacity Q^k is (7.9)–(7.14) with (7.12) and (7.13) replaced by (7.19), (7.20), and:

$$\begin{aligned} \sum_{i \in N} z_i^k &= 1 & \forall k \in K \\ \sum_{k \in K} z_i^k &= x_{0i} & \forall i \in N \\ z_i^k &\in \{0, 1\} & \forall i \in N, k \in K \\ f_{0i} &\leq \sum_{k \in K} Q^k z_i^k & \forall i \in N. \end{aligned}$$

This system becomes inequalities (7.21) when all vehicles have identical capacity Q .

While f_a determines a flow delivering a commodity along the route determined by x_a , the value $Q - f_a$ can be seen as another flow collecting a different product (e.g., the empty space in the vehicle). Each flow delivers or collects q_i units through customer i . Inequality $f_{ij} \leq Q x_{ij}$ forces $f_{ji} = 0$ when $x_{ij} = 1$. Thus, only the delivery flow is reflected in the f variables. Baldacci et al. (1999) proposed replacing constraints (7.12) and (7.13) in the two-index flow formulation (7.9)–(7.14) with:

$$\begin{aligned} \sum_{a \in \delta^-(i)} h_a - \sum_{a \in \delta^+(i)} h_a &= 2q_i & \forall i \in N \\ h_{ij} + h_{ji} &= Q(x_{ij} + x_{ji}) & \forall i, j \in N \mid i < j \\ h_a &\geq 0 & \forall a \in A. \end{aligned}$$

Letchford and Salazar-González (2006) proved that it is equivalent to SCF1a.

Multi-commodity flow formulations: A weakness in the previous compact models is the use of the big value Q to deactivate (7.15)–(7.16) when $x_{ij} = 0$, and (7.22) when $x_{ij} = 1$. Some improvement is made with (7.17)–(7.18) and (7.23), respectively. However, the LP relaxation is further improved by using several flow variables on each arc instead of only one, leading to the so-called *multi-commodity flow formulations*. The price to pay for having a better LP

relaxation is to have more variables in the model, and only computational experiments may determine whether such a price is worth paying.

The first multi-commodity flow formulation was presented in Garvin et al. (1957). It uses a family of continuous variables to model the path from the depot to each customer k . More precisely, for each $k \in N$, let f_a^k be a variable taking the value 1 when a vehicle traverses a on the way from the depot to customer k . The formulation is then obtained by replacing constraints (7.12)–(7.13) in the two-index flow formulation (7.9)–(7.14) with:

$$\sum_{a \in \delta^+(0)} f_a^k = \sum_{a \in \delta^-(k)} f_a^k = 1 \quad \forall k \in N \quad (7.24)$$

$$\sum_{a \in \delta^-(0)} f_a^k = \sum_{a \in \delta^+(k)} f_a^k = 0 \quad \forall k \in N \quad (7.25)$$

$$\sum_{a \in \delta^+(i)} f_a^k = \sum_{a \in \delta^-(i)} f_a^k \quad \forall k, i \in N \mid i \neq k \quad (7.26)$$

$$f_a^k \geq 0 \quad \forall k \in N, a \in A \quad (7.27)$$

and

$$\sum_{k \in N} q_k f_a^k \leq Q x_a \quad \forall a \in A. \quad (7.28)$$

The LP relaxation provides a better bound when adding:

$$f_a^k \leq x_a \quad \forall k \in N, a \in A.$$

The resulting CVRP formulation is called MCF1a. An alternative formulation, called MCF1b, can be obtained by replacing (7.28) with the following constraints:

$$\sum_{k \in N \setminus \{i\}} q_k \sum_{a \in \delta^+(i)} f_a^k \leq Q - q_i \quad \forall i \in N. \quad (7.29)$$

Letchford and Salazar-González (2015) show computational results indicating that no dominance relation between the LP relaxations of MCF1a and MCF1b exists.

For a given arc (i, j) , inequality (7.28) limits to Q the demand of all customers served after i , including j . Since $f_{ij}^j = x_{ij}$, inequalities (7.28) can be rewritten as:

$$\sum_{k \in N \setminus \{j\}} q_k f_{ij}^k \leq (Q - q_j) x_{ij} \quad \forall (i, j) \in A.$$

Note that $f_{ij}^i = 0$ even when $x_{ij} = 1$ thus, the demand of i is not considered in the previous inequality. A strengthening is obtained by also reducing Q with q_i as follows:

$$\sum_{k \in N \setminus \{i, j\}} q_k f_{ij}^k \leq (Q - q_i - q_j) x_{ij} \quad \forall (i, j) \in A. \quad (7.30)$$

The formulation MCF1a with (7.28) replaced by (7.30) is referred to as MCF1c.

Inequalities (7.29) exploit the fact that $\sum_{a \in \delta^+(i)} f_a^k$ is 1 when both i and k are served by the same vehicle, and i precedes k along the route. Then, they can be strengthened by including not only the customers that are served after i but also the customers that are served before i :

$$\sum_{k \in N \setminus \{i\}} q_k \left(\sum_{a \in \delta^+(i)} f_a^k + \sum_{a \in \delta^+(k)} f_a^i \right) \leq Q - q_i \quad \forall i \in N. \quad (7.31)$$

Adding (7.31) to MCF1c leads to a stronger formulation that we refer to as MCF1d.

The partial precedence relation between customers i and k given by $\sum_{a \in \delta^+(i)} f_a^k$ leads to valid inequalities like:

$$\sum_{a \in \delta^+(i)} f_a^k + \sum_{a \in \delta^+(k)} f_a^i \leq 1 \quad \forall i, k \in N \mid i \neq k$$

and

$$\sum_{a \in \delta^+(i)} f_a^j + \sum_{a \in \delta^+(j)} f_a^k \leq 1 + \sum_{a \in \delta^+(i)} f_a^k \quad \forall i, j, k \in N \mid i \neq j, i \neq k, j \neq k.$$

A nice exercise for the reader is to prove that these linear inequalities are redundant in the presence of (7.10), (7.14), and (7.24)–(7.27).

A different multi-commodity flow formulation with *two* commodities per customer and a better LP relaxation was presented in Letchford and Salazar-González (2006). For each arc a and customer k , the variable f_a^k is defined as before. Still, there is now also a binary variable g_a^k , taking the value 1 if and only if a vehicle traverses a on the way from k to the depot. We then replace the constraints (7.12) and (7.13) in the two-index flow formulation (7.9)–(7.14) with the constraints (7.24)–(7.27), together with:

$$\sum_{a \in \delta^+(k)} g_a^k = \sum_{a \in \delta^-(0)} g_a^k = 1 \quad \forall k \in N \quad (7.32)$$

$$\sum_{a \in \delta^-(k)} g_a^k = \sum_{a \in \delta^+(0)} g_a^k = 0 \quad \forall k \in N \quad (7.33)$$

$$\sum_{a \in \delta^+(i)} g_a^k = \sum_{a \in \delta^-(i)} g_a^k \quad \forall k, i \in N \mid i \neq k \quad (7.34)$$

$$g_a^k \geq 0 \quad \forall k \in N, a \in A, \quad (7.35)$$

and

$$\sum_{k \in N \setminus \{i\}} q_k \left(\sum_{a \in \delta^+(k)} f_a^i + \sum_{a \in \delta^-(k)} g_a^i \right) \leq Q - q_i \quad \forall i \in N \quad (7.36)$$

$$f_a^k + g_a^k \leq x_a \quad \forall k \in N, a \in A. \quad (7.37)$$

The new formulation is referred to as MCF2a, and can be strengthened with

$$\sum_{a \in \delta^+(i)} f_a^k = \sum_{a \in \delta^+(k)} g_a^i \quad \forall k, i \in N \mid i \neq k. \quad (7.38)$$

A further strengthening is obtained by adding

$$\sum_{k \in N \setminus \{i, j\}} q_k (f_a^k + g_a^k) \leq (Q - q_i - q_j) x_a \quad \forall a = (i, j) \in A, \quad (7.39)$$

although in this case (7.36) is redundant for the LP relaxation. Replacing (7.36) with (7.38) and (7.39) in MCF2a is stronger and referred to as MCF2b.

We conclude this section with a recent multi-commodity flow formulation proposed in Letchford and Salazar-González (2015) based on a single continuous variable h_{ij}^k taking a positive value when customer k is in a route traversing arc (i, j) or (j, i) , for each $k \in N$ and $(i, j) \in A$. Considering the variables in the previous formulation, $h_{ij}^k = f_{ij}^k + g_{ji}^k$. In general, the positive value for h_{ij}^k is 1; it is 2 when k is alone in a route and $(i, j) = (0, k)$. Then the multi-commodity flow formulation is the formulation (7.9)–(7.14) with constraints (7.12) and (7.13) replaced with:

$$\begin{aligned} \sum_{a \in \delta^+(0)} h_a^k &= \sum_{a \in \delta^-(k)} h_a^k = 2 & \forall k \in N \\ \sum_{a \in \delta^-(0)} h_a^k &= \sum_{a \in \delta^+(k)} h_a^k = 0 & \forall k \in N \\ \sum_{a \in \delta^+(i)} h_a^k &= \sum_{a \in \delta^-(i)} h_a^k & \forall k, i \in N \mid i \neq k \\ h_a^k &\geq 0 & \forall k \in N, a \in A, \end{aligned}$$

and

$$\begin{aligned}
\sum_{k \in N \setminus \{i, j\}} q_k (h_{ij}^k + h_{ji}^k) &\leq (Q - q_i - q_j)(x_{ij} + x_{ji}) & \forall i, j \in N \mid i < j \\
h_{ij}^k + h_{ji}^k &\leq x_{ij} + x_{ji} & \forall k \in N, i, j \in V \mid i < j \\
\sum_{a \in \delta^+(i)} h_a^k &= \sum_{a \in \delta^+(k)} h_a^i & \forall k, i \in N \mid i < k.
\end{aligned}$$

An interesting result from Letchford and Salazar-González (2015) is that the LP relaxation of this formulation, called MCF3, gives the same bound as the LP relaxation of MCF2b when the cost matrix is symmetric.

7.3.2 Cutting-Plane Formulations

This section shows a procedure to replace (7.12) and (7.13) with linear inequalities on x , without the introduction of continuous variables. The price to pay results in an exponential number of inequalities to manage. It is a methodology that has already successfully exploited to formulate the TSP with the so-called subtour elimination constraints:

$$\sum_{a \in \delta^+(S)} x_a \geq 1 \quad \forall S \subseteq N.$$

They are an alternative to writing constraint (7.13), and it can be adapted also to enforce the vehicle capacity limitations in (7.12) by simply lifting the right-hand side from 1 to the minimum number of vehicles needed to serve customers in S . This lifting operation gives the so-called bin-packing constraints:

$$\sum_{a \in \delta^+(S)} x_a \geq r(S) \quad \forall S \subseteq N. \quad (7.40)$$

They stipulate that each cut $(S, V \setminus S)$ defined by a subset $S \subset N$ in the graph G includes at least $r(S)$ arcs in routes, i.e., a minimum number of vehicles needed to serve set S .

When $|S| = 1$ or $S = N$ the constraint (7.40) is a weakened form of the corresponding degree constraint (7.10). Note also that, because of these degree constraints:

$$\sum_{a \in \delta^+(S)} x_a = \sum_{a \in \delta^-(S)} x_a \quad \forall S \subseteq N.$$

In other words, each cut $(S, V \setminus S)$ is traversed the same number of times in both directions. Hence, one can replace $\delta^+(S)$ with $\delta^-(S)$ in (7.40) without affecting the formulation. Furthermore, there is a third popular rewriting of (7.40) that utilizes the degree constraints (7.10). This leads to the well-known *generalized subtour elimination* constraints:

$$\sum_{a \in A(S)} x_a \leq |S| - r(S) \quad \forall S \subseteq N.$$

A major drawback of using the inequalities in (7.40), no matter the opted rewriting, in a formulation is the non-explicit right-hand side, which depends on the solution of a combinatorial optimization problem. However, one can replace $r(S)$ with the trivial BPP bound $\lceil q(S)/Q \rceil$, and the (weaker) linear inequality remains valid. The new inequalities

$$\sum_{a \in \delta^+(S)} x_a \geq \lceil q(S)/Q \rceil \quad \forall S \subseteq N \quad (7.41)$$

are so-called *rounded capacity-cut constraints*, and the CVRP formulation obtained from (7.9)–(7.14) by replacing (7.12) and (7.13) with (7.41) is denoted by RCC.

There is still a second drawback related to the large number of inequalities. With the cardinality growing exponentially with n , it is practically impossible to solve the LP relaxation of these CVRP formulations if one tries to generate *all* the linear inequalities before calling the Mixed Integer Linear Programming (MILP) solver. A possible way to partially overcome this drawback is to consider only a limited subset of the constraints and add a missing one only when needed in an iterative cutting-plane framework. It requires an appropriate procedure to check whether a (possibly fractional) solution x^* of the relaxed model is violating a missing linear inequality of the family and determining a set S associated with a violated constraint, if it exists. This very specific question is called the *separation problem* of the family of inequalities, and a solution algorithm is called the *separation procedure*.

When x^* is an integer vector, the separation problem is easy (i.e., it is solvable in polynomial time). It amounts to finding the connected components of a graph $G^* = (N, A^*)$ where $A^* = \{a \in A : x_a^* = 1\}$. The separation procedure ends by checking the sum of customers' demands in each connected component and defining a violated constraint for each component with a sum of customers' demands larger than Q . With such a simple procedure, one can implement a so-called *cut-and-branch* algorithm for solving the CVRP.

Let us discuss the differences between cut-and-branch and branch-and-cut approaches. Among other details, there are two major difficulties when solving the RCC formulation. One difficulty is the large number of inequalities in (7.40) or (7.41), as mentioned above. This difficulty is addressed in the “cut” phase. The other major difficulty is that we want the decision variable to take integer values, as stated in (7.14). This other difficulty is addressed in the “branch” phase. When designing an algorithm (or implementing a computer code) for solving the CVRP, each difficulty requires considering a set of decisions. A way to address both difficulties is to adopt a specific hierarchy for them: decide on one at an external level and explore the best decision for the other at an internal level. As a consequence, there are two types of algorithmic frameworks: branch-and-cut and cut-and-branch, depending on the phase addressed at each level.

In a branch-and-cut framework, the external level explores a branch-and-bound tree by computing a fractional solution at each branch-and-bound node. Thus, the internal level needs to find a violated inequality (if any exists) by the fractional solution. In a cut-and-branch framework, the branch-and-bound tree is explored internally, which means that the external level is concerned with finding a violated inequality (if any exists) for a given integer solution. While the separation procedures for integer solutions are easier than those for fractional solutions, a major drawback of a cut-and-branch implementation is the enormous effort that the MILP solver must spend to produce invalid CVRP solutions.

Branch-and-cut approaches are (in general) more effective implementations but require efficient separation procedures dealing with fractional solutions x^* . It is quite a difficult task in the CVRP context, especially if we want to keep $r(S)$ on the right-hand side of the capacity constraints, as in (7.40). Even if replacing $r(S)$ by $\lceil q(S)/Q \rceil$ helps, the separation problem for inequalities (7.41) remains difficult. More precisely, Diarrassouba (2017) proved that it is strongly *NP-hard*, even when restricting to CVRP instances with identical customers' demands. This does not preclude implementing an exact separation procedure for inequalities (7.41). We now describe an approach that does not run in polynomial time unless $P = NP$.

Let x^* be a fractional vector. For each $k \in \{1, \dots, m-1\}$, the separation problem is equivalent to checking whether there is a customer subset S requiring more than k vehicles to be served and such that $\sum_{a \in \delta^+(S)} x_a^* < k+1$. If S exists then the violated inequality to add to the master problem is $\sum_{a \in \delta^+(S)} x_a \geq k+1$.

The above problem is equivalent to finding a minimum cut $(S, V \setminus S)$ in a network with the additional requirement that one side of the cut must include customers with a total demand larger than $k \cdot Q$. To present a mathematical formulation for this problem, let us consider a binary variable y_i taking value 1 when customer i belongs to the side satisfying the additional requirement, and 0 otherwise. For convenience of notation, we also use y_0 for the depot and fix it to 0 to keep the depot outside S . We also need a continuous variable w_a for each arc $a \in A$, represented by w_{ij} when $a = (i, j)$. Then an exact separation procedure for inequalities (7.41) requires solving:

$$\begin{aligned} \min \quad & \sum_{a \in A} x_a^* w_a \\ & w_{ij} \geq y_i - y_j \quad \forall (i, j) \in A \\ & w_a \geq 0 \quad \forall a \in A \\ & \sum_{i \in N} q_i y_i \geq k \cdot Q + 1 \\ & y_i \in \{0, 1\} \quad \forall i \in N \\ & y_0 = 0 \end{aligned}$$

for each $k \in \{1, \dots, m-1\}$.

The situation changes drastically when $r(S)$ is replaced by the weaker BPP bound $q(S)/Q$, leading to the so-called *fractional capacity-cut constraints*:

$$\sum_{a \in \delta^+(S)} x_a \geq q(S)/Q \quad \forall S \subseteq N. \quad (7.42)$$

In this case, given a (possibly fractional) solution x^* of a relaxed model, the separation procedure consists of a $s-t$ max-flow computation on the following capacitated network:

- The vertex set includes the depot 0, the customers, and a dummy vertex $n+1$.
- The arc set includes each arc $a \in A$ with $x_a^* > 0$, and the arc $(0, i)$ for each customer i .
- The capacity of an arc from A is x^* , and that of the arcs $(0, i)$ is q_i/Q .

The $s-t$ flow from vertex $s = 0$ to the dummy vertex $t = n + 1$ with maximum capacity must then be computed. Let f^* be the capacity of an optimal flow and let $(\{0\} \cup S, N \setminus S \cup \{n + 1\})$ be the cut in the network defined by an optimal dual solution (thus, a cut separating s and t with minimum capacity). If $f^* \geq q(N)/Q$, the solution x^* satisfies all fractional capacity-cut constraints. Otherwise, the subset S of customers determines a fractional capacity-cut constraint (7.42) violated by x^* . Clearly, it is convenient to round up the right-hand side when the violated constraint is added to the next relaxed model, so the relaxed model is strengthened with a new inequality (7.41). While the described procedure is an exact approach to solving the separation problem of the fractional capacity-cut constraints (7.42), it can also be seen as a heuristic approach to solve the separation problem of the rounded capacity-cut constraints in (7.41). This separation procedure implemented in a *branch-and-cut* framework leads to an effective algorithm for the CVRP.

The branch-and-cut algorithm can be improved in different ways, and one requires separating other families of inequalities. The so-called *generalized large multi-star inequalities* (GLMs) is an example of one of these families. They were introduced by Gouveia (1995) as an alternative to $\lceil q(S)/Q \rceil$ for strengthening the ratio $q(S)/Q$. Indeed, the vehicles serving the customers in a subset S also need to serve the customers immediately routed after or before a customer in S . This means that the numerator of the ratio can be increased by q_j whenever there is an arc (i, j) or (j, i) routed with $i \in S$. The GLM inequalities are then:

$$\sum_{a \in \delta^+(S)} x_a \geq \frac{1}{Q} \sum_{i \in S} \left(q_i + \sum_{j \in N \setminus S} q_j (x_{ij} + x_{ji}) \right) \quad \forall S \subseteq N. \quad (7.43)$$

Like the rounded capacity-cut constraints (7.41), the GLM inequalities (7.43) are stronger than the fractional capacity-cut constraints (7.42). No dominance relation exists between the GLM inequalities and the rounded capacity-cut constraints, with the advantage that the separation problem of the GLM inequalities is still polynomial-time solvable (see Lysgaard et al., 2004). Observe that the exact separation of the GLM inequalities suggests another heuristic separation of the rounded capacity-cut constraints. Furthermore, in an efficient cutting-plane approach for the CVRP, no matter how a potential customer set S is generated, it is worth checking the violation of different inequalities defined by S (including (7.41) and (7.43)) and strengthening the relaxed model with the inequalities violated by the current solution x^* .

We conclude this section by observing that the above formulations with an exponential number of constraints can easily be improved when the travel costs are known to be a symmetric matrix. Indeed, when $c_{ij} = c_{ji}$ for all $i, j \in V$, the customers along a route may be visited indifferently clockwise or counter-clockwise; hence, it is unnecessary to know in which direction arcs are traversed by the vehicles, and each pair $\{(i, j), (j, i)\}$ can be replaced by a single edge $\{i, j\}$, meaning that the CVRP can be modeled on an undirected graph $G = (V, E)$. A clear advantage is that the number of x variables is halved. When single-customer routes are not allowed, the edges incident to the depot can be traversed at most once, like other edges. Instead, when single-customer routes are allowed, an edge connecting the depot and a customer may be traversed twice. To model this minor observation, one can either include in the model both binary variables x_{0i} and x_{i0} , or use a single integer variable x_{0j} , which may take the value $\{0, 1, 2\}$. In this latter case, $x_{0i} = 2$ means that customer i is alone in the route of

a vehicle. The following is a flow formulation for a symmetric CVRP instance where single-customer routes are allowed.

$$\min \sum_{e \in E} c_e x_e \quad (7.44)$$

subject to

$$\sum_{e \in \delta(i)} x_e = 2 \quad \forall i \in N \quad (7.45)$$

$$\sum_{e \in \delta(0)} x_e = 2 \cdot m \quad (7.46)$$

$$\sum_{e \in \delta(S)} x_e \geq 2 \cdot r(S) \quad \forall S \subseteq N \quad (7.47)$$

$$x_e \in \{0, 1\} \quad \forall e \notin \delta(0) \quad (7.48)$$

$$x_e \in \{0, 1, 2\} \quad \forall e \in \delta(0). \quad (7.49)$$

The degree constraints (7.45) impose that exactly two edges are incident to each vertex associated with a customer. Equation (7.46) states that $2m$ edges are incident to the depot vertex, respectively. The capacity-cut constraints (7.47) impose both the connectivity of the solution and the vehicle capacity requirements by forcing a sufficient number of edges to enter each subset of vertices. Also in this case, due to equation (7.45), the capacity-cut constraints (7.47) may be rewritten as the generalized subtour elimination constraints:

$$\sum_{e \in E(S)} x_e \leq |S| - r(S) \quad \forall S \subset N,$$

and again $r(S)$ may be replaced by the trivial BPP bound $\lceil q(S)/Q \rceil$, or even by $q(S)/Q$ to work with linear inequalities that can be separated in polynomial time.

7.3.3 Set Partitioning Formulations

The *Set Partitioning* (SP) formulation of the VRP was originally proposed by Balinski and Quandt (1964) and uses a possibly exponential number of binary variables, each associated with a feasible route for a single vehicle. A route r in G is feasible when it is a circuit in A visiting the depot and a subset S of customers, each one once, such that $\sum_{i \in S} q_i \leq Q$. Let R be the collection of all feasible routes in G . For each $r \in R$, if $i \in V$, let a_i^r be the number of

times that i is visited by r ; if $a \in A$, let b_a^r be the number of times that a is traversed by r . Let c_r be the cost of the route r , which, according to our initial definition of CVRP, means $c_r = \sum_{a \in A} b_a^r c_a$, but other more sophisticated cost functions could be acceptable to state c_r . Consider a binary variable z_r for each $r \in R$ taking value 1 when r is selected in an optimal CVRP solution. The model is:

$$\min \sum_{r \in R} c_r z_r \quad (7.50)$$

subject to

$$\sum_{r \in R} a_i^r z_r = 1 \quad \forall i \in N \quad (7.51)$$

$$\sum_{r \in R} z_r = m \quad (7.52)$$

$$z_r \in \{0, 1\} \quad \forall r \in R. \quad (7.53)$$

Constraints (7.51) impose that each customer i is covered by exactly one of the selected routes, and (7.52) requires that m routes are selected. This is a very general model that may easily consider several constraints, such as time windows, since route feasibility is implicitly considered in the definition of set R .

Observe that if the cost matrix satisfies the triangle inequality, then equation (7.51) can be replaced by “greater than or equal” inequalities, thus leading to a *set covering* model. The main advantage of using the set covering formulation over the set partitioning one is that in the former, only inclusion-maximal feasible circuits, among those with the same cost, need to be considered in the definition of R . This considerably reduces the number of variables. In addition, when using a set covering formulation, the dual solution space is considerably reduced since dual variables are restricted to non-negative values only.

One of the main drawbacks of these models is the exponential number of variables. The explicit generation of all the feasible routes is thus normally impractical, and one has to resort to a *column-generation* approach to solve the LP relaxation of these models. Like a row-generation approach, a column-generation approach intends to solve the complete model by solving a sequence of incomplete (smaller) models. It exploits the fact that an optimal solution only needs a few variables; an optimal CVRP solution only needs m routes. If these few variables (or routes) are in the incomplete model, the solution is optimal, and the iterative approach stops. However, a column-generation approach requires solving a *pricing problem*, which consists of checking whether a missing variable needs to be added to the incomplete problem and proceeding with the next iteration. When solving an LP relaxation, a missing variable has a negative reduced cost. In the CVRP, the pricing problem consists of finding the shortest route in a network. Taking into account that a route is a simple path with a resource

limitation due to the delivered demands and the vehicle capacity, the pricing problem for solving the model (7.50)–(7.53) is strongly NP-hard. Again, it does not preclude that one can solve the pricing problem to optimality. Just to mention a simple exact pricing approach, we now formulate it as a MILP model.

Let us introduce a binary variable v_a representing the number of times that a is in an optimal route of the pricing problem for each $a \in A$. For simplicity of notation, we also introduce another binary variable w_i representing the number of times an optimal route visits i . To avoid subtours disconnected from the depot, let us use a continuous variable representing the position where a customer i is along an optimal route. Given an optimal solution to the incomplete master problem, let c_a^* be the reduced cost of the variable x_a for each $a \in A$. The MILP model for the pricing problem is then:

$$\begin{aligned}
 & \min \sum_{a \in A} c_a^* v_a \\
 & \sum_{a \in \delta^+(i)} v_a = \sum_{a \in \delta^-(i)} v_a = w_i \quad \forall i \in N \\
 & \sum_{a \in \delta^+(0)} v_a = \sum_{a \in \delta^-(0)} v_a = 1 \\
 & \sum_{i \in N} q_i w_i \leq Q \\
 & u_j \geq u_i + v_{ij} - (n-2)(1-v_{ij}) + (n+3)v_{ji} \quad \forall i, j \in N \mid i \neq j \\
 & v_a \in \{0, 1\} \quad \forall a \in A \\
 & w_i \in \{0, 1\} \quad \forall i \in N.
 \end{aligned}$$

An optimal solution (v^*, w^*, u^*) determines a route $r \in R$ with minimum reduced cost $c_r^* = \sum_{a \in A} b_a^r c_a^*$. When $c_r^* < 0$, it suggests a missing variable z_r be added to the incomplete problem and then another iteration follows.

The pricing problem becomes easier if a customer can be visited several times along the path, i.e., if R is extended to include non-elementary routes. Indeed, there are dynamic programming approaches to solving the pricing problem in $O(n^2 Q)$ (see, e.g., Irnich and Desaulniers, 2006). The price to pay for making the pricing problem easier is that the LP relaxation is weaker. Still, several researchers have shown that the advantages compensate for the disadvantages, and larger CVRP instances can be solved.

A further improvement to the model arises from the fact that the x_a variables of the flow formulations can be linked to the z_r variables by simply adding the equations:

$$x_a = \sum_{r \in R} b_a^r z_r \quad \forall a \in A.$$

(Note that a_i^r and b_a^r can be larger than 1 when r is a non-elementary route.) The advantage of including the x variables in the formulation is that the LP relaxation can now be strengthened with other inequalities, such as the rounded capacity constraints.

A major issue is the effect of the dual variables of the additional constraints in the pricing problem. When a family of additional constraints only involves the additional variables x_a and not the original z_r , then the new dual variables do not change the pricing subproblem structure, and the constraints are called robust cuts. A branch-and-cut-and-price approach using robust cuts is described in Fukasawa et al. (2006) and used to solve CVRP instances with up to 134 customers. Baldacci et al. (2008) describes a different approach using non-robust cuts with a very useful suggestion to manage the complexity of the pricing subproblem. It consists of generating all the routes with reduced costs smaller than the gap at the root node of the branch-and-bound search; thus, pricing is unnecessary for other nodes. We recommend Pecin et al. (2017), Costa et al. (2019), and Letchford and Salazar-González (2019) for further references and details.

7.4 GETTING FAMILIAR WITH THE MODELS ON A COMPUTER AND SOME CONCLUSIONS

To help the reader better understand the models, they can find all the mentioned formulations at <https://github.com/jjsalaza/CVRPmodels.git>. It is a single file written in the Julia programming language. It includes 13 compact models and four non-compact formulations described in this chapter. The instances can be selected from the CVRP library, and “E-n13-k4” is the default example (it has $n = 12$, $m = 4$, $Q = 6000$, and the optimal value is 247). The code needs mixed integer linear programming software to solve an instance using a model. By default, the code calls “Gurobi,” but other solvers can easily replace it if desired.

The aim of providing this source code to the reader is only to help compare the presented models and become familiar with the effect of modifying the formulations. No routine in this code is the fastest approach to solving a CVRP instance today. Using this code, the reader can generate (for example) Table 7.1, which shows the percentage of the lower bound from the linear programming relaxation of some models for some small-size instances. From this table, the formulation MCF2b is the one providing the strongest LP relaxation on these CVRP instances. We encourage the reader of this chapter to download and experiment with this source code.

A conclusion after some experiments is that the performance of the solution approach strongly depends on the input parameters that define the CVRP instances. While the difficulties clearly increase with n and m , the BPP component has a direct impact, and the instance is more difficult for a model when packing the customer demands is tighter. Considering a CVRP instance, a model is generally better when the bound given by the LP relaxation is higher. Other considerations affect this conclusion, such as the number of variables and constraints involved in the formulation. While compact formulations can be easily fed into modern solvers, there has been a lot of progress with helping to write variable and constraint generation approaches.

This second contribution is not exclusive to implementing cutting-plane or column-generation approaches, but they should also be considered when implementing a compact formulation as size matters here too. Indeed, even if the number of variables f_a^k used by some formulation is a polynomial number close to n^3 , this number is a million when $n = 100$, and only very few are necessary to determine an optimal CVRP solution. A similar reasoning applies to the constraints $f_a^k \leq x_a$, which are also close to a million inequalities when $n = 100$. Thus, the dynamic generation of variables and constraints should also be evaluated when solving compact formulations. In other words, whether the models rely on a polynomial or

Table 7.1 *Quality of the LP bound for selected benchmark instances provided by some formulations described in this chapter*

name	MTZ1	MTZ2	SCFla	SCFlb	MCF1a	MCF1b	MCF1c	MCF1d	MCF2a	MCF2b
E-n13-k4	90.23	91.5	93.65	96.49	94.64	92.23	96.87	96.98	95.84	97.51
E-n22-k4	74.38	82.27	82.66	88.15	89.67	85.19	93.34	93.74	90.29	95.91
E-n23-k3	82.51	91.18	84.51	86.08	92.88	91.11	94.06	95.01	93.71	96.26
P-n16-k8	90.2	91.23	91.05	94.16	91.34	89.34	94.4	94.49	93.14	95.21
P-n19-k2	80.28	86.79	84.73	86.92	91.61	90.57	91.87	91.87	91.71	92.09
P-n20-k2	83.83	90.28	88.13	90.05	94.21	91.44	95.19	95.3	93.69	95.97

exponential number of variables and constraints, appropriate management of them through pricing and separation procedures, respectively, is crucial to reducing the computational time consumed by the solver.

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